

DOCUMENT 3: PROBLEM STATEMENT & PROPOSED SOLUTION

Project Name: PARIKSHAK-AI (परीक्षक-AI)

Challenge: Challenge 03 — AI-Driven Examination & On-Screen Marking Transformation

Track: Technical Track | **Team Name:** eMitra

1. Comprehensive Problem Statement: The Examination Crisis in Madhya Pradesh

Higher education in Madhya Pradesh represents one of India's most expansive state university frameworks. According to AISHE data, the state accommodates over **28.0 Lakh enrolled students** across **2,610+ affiliated colleges** and **~94 universities** (25 State Public Universities, 55 State Private Universities, 2 Central Universities, and specialized institutes).

Every academic year entails two primary examination cycles (Odd Semesters: Nov–Jan; Even Semesters: May–July), producing an overwhelming aggregate of **3.0 to 3.4 Crore handwritten answer booklets** (averaging 32 to 40 pages per booklet).

The Structural Vulnerabilities of the Current Examination Lifecycle

A. The Collapse of Outsourced Manual & Vendor Workflows (The BU Bhopal Debacle)

Barkatullah University (BU), Bhopal faced an administrative crisis when results processed by its outsourced result-processing agency showed suspected discrepancies in around **30% of released results**. Nearly 10,000 complaints reached the university, which sought an explanation from the agency:

- Students who sat the exams were marked **"ABSENT"**, given zero marks, or shown subjects they had not opted for.
- At **DAVV Indore**, only about 43% of ~7,200 B.Ed candidates passed a first-semester exam; after allegations of faulty evaluation the university ordered an expert re-evaluation of answer sheets.

B. The Severe Operational Economics of Faculty Valuation

State universities empanel between **15,000 and 25,000 college and university teachers** per evaluation window. Under ordinances such as DAVV Ordinance No. 5 and RGPV statutory guidelines:

- Evaluator compensation is pegged at an austere **₹12 to ₹18 per booklet** for undergraduate humanities/commerce and **₹20 to ₹25 per booklet** for engineering and professional courses.
- An examiner spending 4–5 hours evaluating 40 answer books earns approximately ₹600 to ₹800—often paid after 6 to 12 months of bureaucratic delays.
- A standard booklet contains 32 to 36 pages of complex handwritten prose, mathematical derivations, or architectural schematics. Evaluators are pressured to clear daily quotas of 40 to 60 books within a 3 to 4 hour evaluation center shift.
- **The Mathematical Impossibility:** Evaluating a 36-page script in 180 to 240 seconds allows barely **5 to 7 seconds per page**. Evaluators cannot cognitively read the script. They resort to **"glance-checking"**—evaluating based on penmanship, paragraph length, and apparent neatness. By 3:00 PM, evaluator cognitive fatigue leads to severe central-tendency bias (assigning uniform passing scores of 12–15 out of 20 to every candidate).

C. Systemic Result Backlogs & Career Disruption

University ordinances mandate result declarations within 30 to 45 days of the final examination. In reality, examination branches at BU Bhopal, Jiwaji University Gwalior, RDVV Jabalpur, and APSU Rewa report result

backlogs spanning **3 to 14 months**.

- Final-year students miss corporate onboarding cut-offs at IT services and core manufacturing firms (which strictly mandate zero active backlogs at joining).
- Candidates are barred from state public service examinations (MPPSC State Services) or national tests (UPSC, Banking, GATE) due to unavailable provisional degree certificates.

D. The Predatory Revaluation Industry

Because first-tier valuation is plagued by glance-checking and transcription blunders, an estimated **6% to 10% of candidates apply for revaluation and challenge evaluation**. Universities in MP charge between **₹500 and ₹1,500 per subject** for revaluation and copy inspection. Across 7 major state universities (DAVV, BU, RGPV, Jiwaji, Vikram, RDVV, APSU), an estimated **6 Lakh to 9 Lakh revaluation requests** are processed annually, generating **₹30 to ₹45 Crore in non-refundable fee revenue**. Large mark changes on revaluation point to unreliable first-tier evaluation.

E. Security Paralysis & The Shift from L1 to QCBS Post-CBSE OSM 2026

The national CBSE On-Screen Marking controversy in 2026—where low-cost vendor Coempt Edu Teck’s platform failed under an aggressive L1 bid (₹25.74 vs TCS’s ₹63.4/script), exposing critical Broken Object Level Authorization (BOLA/IDOR) vulnerabilities and inverted candidate rolls—sent shockwaves through MP’s higher education leadership. In early 2026, **DAVV Indore placed its planned ₹2.0 Cr digital evaluation rollout under scrutiny**, requiring a strict 10-day physical pilot testing answer booklet spine cutting, leaf scanning, and data custody before awarding tenders. Simultaneously, active digital evaluation tenders across MP—including **BU Bhopal (Tender ID: 2026_BU_504007_1)**, **Jiwaji University (Tender ID: 2026_JIWAJ_509311_1)**, and **Vikram University (NIT Feb 24, 2026)** on `mptenders.gov.in`—demand state-sovereign, secure, and auditable digital evaluation architectures.

The 4 Core Stakeholder Personas & Target User Mapping

Stakeholder Persona	The Operational Ground Reality & Pain Point	How PARIKSHAK-AI Solves It
Persona 1: The University Examiner (Evaluator)	Underpaid (₹16–₹25/booklet), forced to glance-check 40 scripts of 36 pages in 3 hours, suffering severe cognitive fatigue by 3:00 PM with zero rubric assistance.	Intelligent Cognitive Copilot: Pre-reads Hindi/English script via multimodal vision, pre-populates rubric breakdown; examiner retains 100% authority via 1-click Accept/Modify, reducing the reading burden.
Persona 2: The Controller of Examinations (CoE)	Zero in-flight visibility; reliant on random 5% post-facto sampling; defenseless against High Court contempt notices and results delayed by 3–14 months.	Live State Command Center & In-Flight Calibration: Injects blind seed scripts to detect evaluator drift in real-time; tracks velocity heatmaps; generates predictive result completion dates.
Persona 3: The Student & Citizen	High-stakes career disruption; lost corporate recruitment cut-offs (TCS, Infosys) due to result delays; ₹500–₹1,500 "revaluation tax" on rural/poor students.	Fair, Fast & Transparent Results: Results declared in 15–20 days; eliminates clerical transcription/totaling errors; provides NEP 2020 diagnostic feedback cards mapping conceptual strengths.
Persona 4: MPOnline & Higher Education Department	Institutional trust erosion; managing fragmented, risky private vendor tenders; needing citizen-centric digital public infrastructure.	Citizen-Centric Governance: Sovereign cloud deployment at MP State Data Centre (SDC); non-invasive bolt-on to MPOnline ASP.NET/Oracle; compliant with MeitY MeghRaj and Section 63 BSA 2023.

2. The Proposed Solution: PARIKSHAK-AI (परीक्षक-AI)

PARIKSHAK-AI is a cloud-native, Human-in-the-Loop (HITL) Cognitive Evaluation Platform specifically engineered to plug into MPOnline’s state examination portal. It delivers **citizen-centric governance** in higher education not

by attempting to replace university professors, but by providing an intelligent cognitive assistant that shields examiners from burnout, standardizes grading rubrics, prevents speed-checking anomalies, and produces ironclad legal documentation.

PARIKSHAK-AI CORE WORKFLOW

[Step 1: Scan & Anonymize]

Industrial Scanner → MP SDC Secure Ingestion → Fictitious Code Mapping

[Step 2: Cognitive Pre-Evaluation]

Multimodal Vision Model (Gemini Flash) ingests raw scan + marking rubric. Extracts student textual/diagrammatic evidence; pre-populates marks.

[Step 3: Examiner Interactive Evaluation]

Faculty logs in via 2FA; views scan alongside AI rubric suggestions. Annotates using digital stylus/touchpad; 1-click Accept / Modify / Override

[Step 4: Autonomous Quality Assurance]

• Blind Seed Script Injection (Calibrates examiner consistency)

• Dwell-Time Velocity Sentinel (prompts on rushed marking, no lock)

• Shannon Entropy Monitor (Flags uniform rubber-stamp grading)

[Step 5: Psychometric Forensics & Tabulation]

• Live item analysis (point-biserial) question ambiguity detection

• Direct API push to MPOnline Tabulation Register

• 1-Click Cryptographic RTI Justification Dossier Generation

3. Direct Mapping to Challenge 03 Requirements

The table below demonstrates how PARIKSHAK-AI directly satisfies every single feature specified in the MPOnline Idea Hackathon Challenge 03 mandate:

#	Challenge 03 Requirement	Current Gap in MP Universities	PARIKSHAK-AI Solution & Technical Approach
1	AI-assisted answer evaluation support	Manual glance-checking in 180s; no marking guidance.	Decomposed Analytic Rubric Engine provides question-by-question scoring recommendations with quoted textual justifications.
2	Automated detection of unchecked answers / marking anomalies	Unchecked pages cause thousands of failing grades annually.	Computer vision page traversal detector verifies all question blocks are assessed; enforces mandatory blank-page watermarking.
3	Examiner performance analytics	Zero visibility into evaluator pacing or fatigue.	Real-time telemetry tracking; scripts evaluated per hour, variance from cohort average, dwell-time distribution across booklet pages.
4	Smart moderation workflows	Arbitrary 5% manual sampling by Head Examiners.	Blind Seed-Script Injection: Invisibly inserts pre-calibrated anchor scripts; flags evaluator drift if deviation exceeds ±15%.
5	Handwriting recognition assistance	60%+ Hindi scripts illegible to standard OCR engines.	End-to-end multimodal vision reasoning processes cursive Devanagari, English, and Hinglish directly from raw imagery.
6	AI-generated evaluation summaries	Marksheets display single numbers with zero diagnostic feedback.	Synthesizes individual student diagnostic reports mapping strengths and conceptual misconceptions aligned with NEP 2020 OBE rubrics.

#	Challenge 03 Requirement	Current Gap in MP Universities	PARIKSHAK-AI Solution & Technical Approach
7	Real-time evaluation dashboards	Controllers of Examination (CoEs) rely on weekly phone calls.	Live State Command Center displays subject-wise completion rates, bottleneck alerts, examiner speed heatmaps, and projected result dates.
8	Malpractice / unusual scoring pattern detection	Unscrupulous examiners award identical passing marks (14/20).	Shannon Entropy Engine: Computes score dispersion entropy ($H(X) < 1.5$); Isolation Forest flags outlier examiner cohorts.
9	Faster result processing	Manual transcription from paper award lists takes 2–4 months.	Zero-touch API ingestion into MPOnline database immediately upon final examiner submission; instant auto-totalling.
10	Mobile-enabled examiner interface	Evaluators bound to obsolete desktop PCs in crowded halls.	Responsive Progressive Web App (PWA) with native stylus and touch support, in line with the June 2026 MP High Court recommendation on stylus marking.

4. Key Differentiators: Why This Solves the Ground Problem

- 1. Non-Invasive Architecture:** We do not propose replacing MPOnline's existing investment in ASP.NET / Oracle ERP. PARIKSHAK-AI acts as a stateless intelligence microservice that ingests scanned images and outputs validated marks.
- 2. Respect for Legal Realities:** Indian higher education law requires degree conferral to rest upon human academic judgment. By maintaining strict Human-in-the-Loop workflows, our solution keeps legal accountability with the examiner while reducing the reading burden.
- 3. Economical Viability:** At an estimated ₹0.5–₹3 per booklet in AI cost, the system is cheap relative to paper logistics, valuation-centre allowances and revaluation administration.